

Yatin Dandi

SOPHOMORE · COMPUTER SCIENCE AND ENGINEERING

IIT Kanpur

☎ (+91) 900-574-0313 | ✉ yatind@iitk.ac.in | 🏠 yatindandi.github.io | 📷 [yatindandi](#)

Education

Indian Institute of Technology Kanpur

BACHELOR OF TECHNOLOGY, MAJOR IN COMPUTER SCIENCE AND ENGINEERING

• Cumulative Performance Index : 10.0/10.0

Kanpur, India

July, 2017 - Present

Interests

Deep Learning, Reinforcement Learning, Computational Neuroscience, Statistical Learning Theory.

Honors & Awards

- 2017 **Aditya Birla Scholarship**, Awarded to 15 students from all IITs
- 2017 **Academic Excellence Award**, Awarded for exceptional performance
- 2016 **KVPY Scholarship, 2016**, Indian Institute of Science
- 2015 **NTSE Scholarship**, Government of India
- 2017 **All India Rank 135**, JEE Advanced 2017
- 2016 **Selected for Indian National Physics Olympiad**, HBCSE
- 2016 **Selected for Indian National Chemistry Olympiad**, HBCSE

Mumbai, India

IIT Kanpur

Bangalore

India

India

Mumbai, India

Mumbai, India

Experience

New York Office, IIT Kanpur

IIT Kanpur

MACHINE LEARNING FOR LARGE SCALE LOGISTICS PLATFORM, UNDER PROF. MANINDRA AGARWAL

May 2018 - July 2018

- Implemented a state of the art algorithm for online collaborative filtering based on Fast Matrix Factorization for Online Recommendation with Implicit Feedback (He et al.) using Numpy and improved the model with sentiment and frequency dependent weighting schemes.
- Used Kafka for real-time data processing and simulated interactions using locust.
- Implemented a recommender system based on deep autoencoders and compared the results with other models using metrics such as hit ratio.
- Implemented a Bidirectional LSTM model using Keras for sentiment analysis of user comments.
- Trained the Latent Dirichlet allocation model on Wikipedia articles for automatic extraction of topics.

Disentangled Representation Learning using Generative Models

IIT Kanpur

UNDERGRADUATE PROJECT UNDER PROF. NISHEETH SRIVASTAVA

May 2018 - Present

- Studied various approaches for learning disentangled representations of sequential data such as using new adversarial loss terms, factorized hierarchical priors and exploiting the probabilistic model and architecture of the LSTM based autoencoder to promote disentanglement.
- Implemented a Variational Autoencoder model for disentangling of time invariant content and dynamics in sequential data (Mandt et al.) using Pytorch.
- Presently implementing an architecture combining VAE and GAN (Larsen et al.) to measure similarities using learned representations and modifying the Disentangled Sequential Autoencoder with hierarchical priors.
- Working on applications of disentangled representations such as transfer in reinforcement learning and video generation.

Social Situation Inference in Simple Animated Shapes

IIT Kanpur

UNDERGRADUATE PROJECT UNDER PROF. NISHEETH SRIVASTAVA

May 2018 - July 2018

- The aim was to characterise the nature and granularity of information about social interactions accessible to human observers from simple visual displays.
- Developed an animation engine capable of producing a vast range of social situations involving autonomous agents having long term and short term goals.
- The physical parameters were inspired by the valence-arousal model and were designed to represent independent factors, together covering the entire emotion space.
- The motion of autonomous characters was based on work by Reynolds, C. W. and the social force model for crowd behavior simulation.
- Currently running experiments with human subjects to determine how subjective attribution of emotions to specific situations varies as a function of the generative models parameters.

Deep Reinforcement Learning for Atari Games

IIT Kanpur

ASSOCIATION FOR COMPUTING ACTIVITIES, IIT KANPUR

February 2018 - May 2018

- Used Numpy to implement various reinforcement learning algorithms such as Dynamic Programming (Policy and Value iteration), Monte Carlo (Epsilon-greedy and off-policy), TD Learning (Q-Learning and SARSA) and Q-Learning with Function Approximation.
- Implemented Deep Q-Learning and Policy Gradient methods for Atari Games using PyTorch and OpenAI Gym.

Image Captioning with Visual Attention

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PROGRAMMING CLUB, IIT KANPUR

February 2018 - May 2018

- Compared various CNN based models for image classification and implemented them using eager execution in Tensorflow.
- Implemented the model described in Show, Attend and Tell (Xu et al.2015) using Tensorflow's estimator API and evaluated the model on MS COCO dataset.

Microsoft code.fun.do hackathon

IIT Kanpur

NATIONAL-LEVEL HACKATHON WINNER

February 2018

- Selected out of 120 students to represent IIT Kanpur at Microsoft Hyderabad center to showcase our project in their academia-industry collaboration event AXLE.
- Built an interactive interface using D3.js to display changing geopolitical relations and popularity of world leaders.
- Used Scrapy, a scraping framework to extract world news.
- Performed sentiment analysis and named entity-recognition on the extracted news to infer the effect of the concerned news on world politics.

POSITIONS OF RESPONSIBILITY

Programming Club

IIT Kanpur

SECRETARY

April 2018 - Present

- Helped conduct and organize Linux fest, introductory workshops and various hackathons.

Relevant Courses

Introduction to Programming - A

Probability for Computer Science - A*

Calculus and Real Analysis - A*

Discrete Mathematics - A

Logic for Computer Science - A*

Computational Cognitive Science (audited)

Linear Algebra and ODE - A

A*: grade for exceptional performance

Skills

Programming Languages

C, C++, Python, Javascript, MATLAB, Octave

Libraries and frameworks

Tensorflow, Pytorch, Scikit-Learn, Pillow, Keras, Numpy

Web

Flask, HTML, CSS, jQuery, MySQL

Utilities

Linux shell utilities, LaTeX, Git

Extra Curricular

- Delivered a talk on Brain-Computer Interface in an event organized by Science Coffee House, IIT Kanpur.
- Winner of Blackbox - a three hour high speed hackathon based on an Esoteric language organized by Programming Club IIT Kanpur.